

AI Exploratory Test Report

Target Application: https://xploresyte.lovable.app

Report Date: March 17, 2026 at 10:40

Tool Version: AI Exploratory Test Engine v2.0

Pages Analyzed: 9

Total Test Cases Generated: 54

Valid Test Cases: 54

Rejected (Hallucinated): 0

Tests Executed: 54

Passed: 32

Failed (Bugs Found): 22

2. Execution Results Summary

32 PASSED

Tests behaved correctly

22 FAILED

Bugs found in application

Bugs Found - Failed Test Cases:

Test ID	Goal	Expected (Not Met - Bug)
VT-01	Break the login form with a long string	Error message for invalid password length
VT-02	Poison the contact form with special characters	Error message for invalid input
VT-05	Test the contact form with empty required fields	Error message for required field
VT-07	Poison the login form with whitespace-only input	Error message for invalid input
VT-09	Break the contact form with a very long phone number	Error message for invalid phone number
VT-11	Test state transition on contact page	Form fields retain values
VT-14	Test calculation update on login page	Error message is updated
VT-16	Explore result update on products page	Order summary is updated
VT-17	Test calculation update on register page	Registration is successful
VT-18	Break the app by entering minimum allowed value in email	Error message for invalid email
VT-19	Break the app by entering maximum allowed value in password	Error message for password length
VT-22	Break the app by entering zero in phone number field	Error message for invalid phone number

VT-23	Break the app by entering a very large number in phone	Error message for invalid phone number
VT-24	Break the app by entering a very small decimal in email	Error message for invalid email
VT-25	Break the app by entering minimum allowed value in pass	Error message for password length
VT-29	Verify contact form retention	Phone number retained in contact form
VT-31	Test login interruption	Login fails due to incomplete form
VT-44	Verify mode switch retains values on dashboard page	Values are retained after mode switch
VT-45	Trigger error on contact form submission with empty fie	Error message is displayed
VT-46	Verify error recovery on contact form with invalid emai	Error message is cleared and form submits suc
VT-50	Test login form error handling with empty password fiel	Error message is displayed for empty password
VT-53	Test contact form error handling with empty name field	Error message is displayed for empty name fie

3. Application Coverage Summary

Metric	Value
Pages Crawled	9
Pages with Interactive Elements	9
Input Fields Identified	9
Buttons Identified	18
Navigation Links	65
Dropdowns/Selects	0
Test Cases Generated	54
Validation Pass Rate	100.0%

4. Batch Generation Strategy

Tests were generated in 6 adversarial batches. Each batch focused on a distinct exploration category to guarantee structural diversity and maximize bug-finding potential.

Batch	Category	Focus
1	Input Torture and Field Poisoning	Edge-case values: long strings, special chars, min-1/max+1, empty required fields.
2	State Transitions and Form Abandonment	Partial fills, navigate-away-return, recalculate after change, state persistence.
3	Boundary Value Analysis	Exact min, max, zero, +1/-1 boundaries across every numeric input field.
4	Multi-Step Navigation Chaos	Fill on page A, jump to B, return to A -- does state survive?
5	Mode Switching and UI State Dances	Toggle modes mid-fill, switch tabs, check value retention after mode change.
6	Error Recovery and Validation Resilience	Submit empty, fix one field, resubmit; clear after success; recovery flow testing.

5. Test Coverage Classification

Category	Count
Authentication Tests	20
General Exploratory Tests	14
Boundary Tests	8
State Persistence Tests	6
Negative Input Tests	4
Validation Enforcement Tests	2

Test cases are automatically classified based on their goals and step patterns. Categories are not mutually exclusive; each test is assigned to its primary category.

6. Validated Test Cases

54 test cases passed anti-hallucination validation. All referenced UI elements exist in the application snapshot.

Each test is marked PASSED (green) or FAILED (red) based on execution results.

VT-01 | Break the login form with a long string [FAILED - BUG FOUND]

Page: /login

Target: Enter password

Type: Boundary Tests

1. Navigate to '/login'
2. Enter '99999999999999999999' into 'Enter password'
3. Click 'Sign In'

Expected: Error message for invalid password length

!! Application did not meet expected behavior - potential bug detected.

VT-02 | Poison the contact form with special characters [FAILED - BUG FOUND]

Page: /contact

Target: Your name

Type: Negative Input Tests

1. Navigate to '/contact'
2. Enter '!@#\$%^&*)' into 'Your name'
3. Click 'Send Message'

Expected: Error message for invalid input

!! Application did not meet expected behavior - potential bug detected.

VT-03 | Exceed the max limit in the register form [PASSED]

Page: /register

Target: Jane Doe

Type: Boundary Tests

1. Navigate to '/register'
2. Enter 'a.repeat(100)' into 'Jane Doe'
3. Click 'Create Account'

Expected: Error message for input length exceeded

VT-04 | Contaminate the register form with cross-field input [PASSED]

Page: /register

Target: Jane Doe

Type: Authentication Tests

- 1. Navigate to '/register'
- 2. Enter 'you@example.com' into 'Jane Doe'
- 3. Click 'Create Account'

Expected: Error message for invalid input

VT-05 | Test the contact form with empty required fields [FAILED - BUG FOUND]

Page: /contact

Target: Your name

Type: Validation Enforcement Tests

- 1. Navigate to '/contact'
- 2. Enter " into 'Your name'
- 3. Click 'Send Message'

Expected: Error message for required field

!! Application did not meet expected behavior - potential bug detected.

VT-06 | Break the products page with rapid button clicks [PASSED]

Page: /products

Target: N/A

Type: General Exploratory Tests

- 1. Navigate to '/products'
- 2. Click 'Add Widget Pro'
- 3. Click 'Add Gadget Lite'
- 4. Click 'Add Component X'

Expected: Error message for rapid button clicks

VT-07 | Poison the login form with whitespace-only input [FAILED - BUG FOUND]

Page: /login

Target: Enter password

Type: Authentication Tests

- 1. Navigate to '/login'
- 2. Enter ' ' into 'Enter password'
- 3. Click 'Sign In'

Expected: Error message for invalid input

!! Application did not meet expected behavior - potential bug detected.

VT-08 | Test the register form with min limit - 1 [PASSED]

Page: /register

Target: Min 8 characters

Type: Boundary Tests

- 1. Navigate to '/register'
- 2. Enter 'a'.repeat(7) into 'Min 8 characters'
- 3. Click 'Create Account'

Expected: Error message for input length too short

VT-09 | Break the contact form with a very long phone number [FAILED - BUG FOUND]

Page: /contact

Target: +1 (555) 000-0000

Type: General Exploratory Tests

- 1. Navigate to '/contact'
- 2. Enter '123456789012345678901234567890' into '+1 (555) 000-0000'
- 3. Click 'Send Message'

Expected: Error message for invalid phone number

!! Application did not meet expected behavior - potential bug detected.

VT-10 | Break form persistence on register page [PASSED]

Page: /register	Target: Jane Doe	Type: State Persistence Tests
1. Navigate to '/register' 2. Enter 'John Doe' into 'Jane Doe' 3. Enter 'johndoe@example.com' into 'you@example.com' 4. Navigate to '/login' 5. Navigate back to '/register'		
Expected: Form fields are empty		

VT-11 | Test state transition on contact page [FAILED - BUG FOUND]

Page: /contact	Target: Your name	Type: State Persistence Tests
1. Navigate to '/contact' 2. Enter 'John Doe' into 'Your name' 3. Enter 'johndoe@example.com' into 'you@example.com' 4. Click 'Send Message' 5. Navigate to '/about' 6. Navigate back to '/contact'		
Expected: Form fields retain values		
!! Application did not meet expected behavior - potential bug detected.		

VT-12 | Explore form abandonment on products page [PASSED]

Page: /products	Target: N/A	Type: General Exploratory Tests
1. Navigate to '/products' 2. Click 'Add Widget Pro' 3. Click 'Add Gadget Lite' 4. Navigate to '/settings' 5. Navigate back to '/products'		
Expected: Previously added items are removed		

VT-13 | Investigate state persistence on dashboard page [PASSED]

Page: /dashboard	Target: N/A	Type: State Persistence Tests
1. Navigate to '/dashboard' 2. Click 'Account v' 3. Navigate to '/contact' 4. Navigate back to '/dashboard'		
Expected: Account menu is closed		

VT-14 | Test calculation update on login page [FAILED - BUG FOUND]

Page: /login	Target: Enter password	Type: Authentication Tests
1. Navigate to '/login' 2. Enter 'wrongpassword' into 'Enter password' 3. Click 'Sign In' 4. Enter 'newpassword' into 'Enter password' 5. Click 'Sign In'		
Expected: Error message is updated		
!! Application did not meet expected behavior - potential bug detected.		

VT-15 | Break registration flow with interrupted user input [PASSED]

Page: /register

Target: Jane Doe

Type: Boundary Tests

1. Navigate to '/register'
2. Enter 'Jane' into 'Jane Doe'
3. Enter 'janedoe@example.com' into 'you@example.com'
4. Navigate to '/login'
5. Navigate back to '/register'
6. Enter 'janedoe123' into 'Min 8 characters'

*Expected: Registration form is partially filled***VT-16 | Explore result update on products page [FAILED - BUG FOUND]**

Page: /products

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/products'
2. Click 'Add Widget Pro'
3. Click 'Order'
4. Click 'Add Gadget Lite'
5. Click 'Order'

*Expected: Order summary is updated***!! Application did not meet expected behavior - potential bug detected.****VT-17 | Test calculation update on register page [FAILED - BUG FOUND]**

Page: /register

Target: Jane Doe

Type: Boundary Tests

1. Navigate to '/register'
2. Enter 'Jane Doe' into 'Jane Doe'
3. Enter 'janedoe@example.com' into 'you@example.com'
4. Enter 'janedoe123' into 'Min 8 characters'
5. Enter 'janedoe123' into 'Repeat password'
6. Click 'Create Account'

*Expected: Registration is successful***!! Application did not meet expected behavior - potential bug detected.****VT-18 | Break the app by entering minimum allowed value in email fie [FAILED - BUG FOUND]**

Page: /login

Target: you@example.com

Type: Authentication Tests

1. Navigate to '/login'
2. Enter 'a@b.c' into 'you@example.com'
3. Click 'Sign In'

*Expected: Error message for invalid email***!! Application did not meet expected behavior - potential bug detected.****VT-19 | Break the app by entering maximum allowed value in password [FAILED - BUG FOUND]**

Page: /login

Target: Enter password

Type: Authentication Tests

1. Navigate to '/login'
2. Enter 'a'.repeat(100) into 'Enter password'
3. Click 'Sign In'

*Expected: Error message for password length***!! Application did not meet expected behavior - potential bug detected.**

VT-20 | Break the app by entering one below minimum allowed value in [PASSED]

Page: /register

Target: Jane Doe

Type: Authentication Tests

1. Navigate to '/register'
2. Enter '' into 'Jane Doe'
3. Click 'Create Account'

*Expected: Error message for empty name***VT-21 | Break the app by entering one above maximum allowed value in [PASSED]**

Page: /register

Target: Repeat password

Type: Authentication Tests

1. Navigate to '/register'
2. Enter 'a'.repeat(101) into 'Repeat password'
3. Click 'Create Account'

*Expected: Error message for password length***VT-22 | Break the app by entering zero in phone number field [FAILED - BUG FOUND]**

Page: /contact

Target: +1 (555) 000-0000

Type: Boundary Tests

1. Navigate to '/contact'
2. Enter '0' into '+1 (555) 000-0000'
3. Click 'Send Message'

*Expected: Error message for invalid phone number***!! Application did not meet expected behavior - potential bug detected.****VT-23 | Break the app by entering a very large number in phone number [FAILED - BUG FOUND]**

Page: /contact

Target: +1 (555) 000-0000

Type: Boundary Tests

1. Navigate to '/contact'
2. Enter '9999999999' into '+1 (555) 000-0000'
3. Click 'Send Message'

*Expected: Error message for invalid phone number***!! Application did not meet expected behavior - potential bug detected.****VT-24 | Break the app by entering a very small decimal in email field [FAILED - BUG FOUND]**

Page: /login

Target: you@example.com

Type: Boundary Tests

1. Navigate to '/login'
2. Enter '0.0001' into 'you@example.com'
3. Click 'Sign In'

*Expected: Error message for invalid email***!! Application did not meet expected behavior - potential bug detected.****VT-25 | Break the app by entering minimum allowed value in password [FAILED - BUG FOUND]**

Page: /login

Target: Enter password

Type: Authentication Tests

1. Navigate to '/login'
2. Enter 'a' into 'Enter password'
3. Click 'Sign In'

*Expected: Error message for password length***!! Application did not meet expected behavior - potential bug detected.**

VT-26 | Break the app by entering maximum allowed value in name fiel [PASSED]

Page: /register

Target: Jane Doe

Type: Authentication Tests

1. Navigate to '/register'
2. Enter 'a'.repeat(50) into 'Jane Doe'
3. Click 'Create Account'

Expected: Error message for name length

VT-27 | Break navigation between dashboard and products [PASSED]

Page: /dashboard

Target: N/A

Type: Authentication Tests

1. Navigate to '/dashboard'
2. Click 'Account v'
3. Navigate to '/products'
4. Click 'Add Widget Pro'

Expected: Widget Pro added to order

VT-28 | Test registration flow interruption [PASSED]

Page: /register

Target: Jane Doe

Type: Authentication Tests

1. Navigate to '/register'
2. Enter 'Jane Doe' into 'Jane Doe'
3. Navigate to '/about'
4. Navigate to '/register'
5. Click 'Create Account'

Expected: Registration fails due to incomplete form

VT-29 | Verify contact form retention [FAILED - BUG FOUND]

Page: /contact

Target: +1 (555) 000-0000

Type: Negative Input Tests

1. Navigate to '/contact'
2. Enter '+1 (555) 000-0000' into '+1 (555) 000-0000'
3. Navigate to '/products'
4. Navigate to '/contact'

Expected: Phone number retained in contact form

!! Application did not meet expected behavior - potential bug detected.

VT-30 | Break settings navigation [PASSED]

Page: /settings

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/settings'
2. Click 'Dismiss'
3. Navigate to '/dashboard'
4. Navigate to '/settings'

Expected: Settings page loads without errors

VT-31 | Test login interruption [FAILED - BUG FOUND]

Page: /login

Target: you@example.com

Type: Authentication Tests

1. Navigate to '/login'
2. Enter 'you@example.com' into 'you@example.com'
3. Navigate to '/about'
4. Navigate to '/login'

5. Click 'Sign In'

Expected: Login fails due to incomplete form

!! Application did not meet expected behavior - potential bug detected.

VT-32 | Verify dashboard navigation [PASSED]

Page: /dashboard

Target: N/A

Type: Authentication Tests

1. Navigate to '/dashboard'
2. Click 'Account v'
3. Navigate to '/contact'
4. Navigate to '/dashboard'

Expected: Dashboard page loads without errors

VT-33 | Break products navigation [PASSED]

Page: /products

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/products'
2. Click 'Add Component X'
3. Navigate to '/settings'
4. Navigate to '/products'

Expected: Component X added to order

VT-34 | Test registration navigation [PASSED]

Page: /register

Target: you@example.com

Type: Authentication Tests

1. Navigate to '/register'
2. Enter 'you@example.com' into 'you@example.com'
3. Navigate to '/login'
4. Navigate to '/register'

Expected: Registration page loads without errors

VT-35 | Verify about page navigation [PASSED]

Page: /about

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/about'
2. Navigate to '/contact'
3. Navigate to '/about'

Expected: About page loads without errors

VT-36 | Break mode switching on products page [PASSED]

Page: /products

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/products'
2. Click 'Add Widget Pro'
3. Click 'Add Gadget Lite'
4. Click 'Show Special Offers'
5. Click 'Add Component X'

Expected: UI does not flicker or crash

VT-37 | Test rapid mode switching on dashboard page [PASSED]

Page: /dashboard	Target: N/A	Type: Authentication Tests
1. Navigate to '/dashboard' 2. Click 'Account v' 3. Click 'Dismiss' 4. Click 'Account v' 5. Click 'Dismiss'		

Expected: No UI errors or inconsistencies

VT-38 | Verify mode switch retains values on contact page [PASSED]

Page: /contact	Target: Your name	Type: State Persistence Tests
1. Navigate to '/contact' 2. Enter 'John Doe' into 'Your name' 3. Enter 'john@example.com' into 'you@example.com' 4. Click 'Dismiss' 5. Click 'Send Message'		

Expected: Values are retained after mode switch

VT-39 | Check for race conditions on products page [PASSED]

Page: /products	Target: N/A	Type: General Exploratory Tests
1. Navigate to '/products' 2. Click 'Add Widget Pro' 3. Click 'Add Gadget Lite' 4. Click 'Add Component X' 5. Click 'Order'		

Expected: No race conditions or UI freezes

VT-40 | Test mode switch with incomplete fields on register page [PASSED]

Page: /register	Target: Jane Doe	Type: Authentication Tests
1. Navigate to '/register' 2. Enter 'Jane' into 'Jane Doe' 3. Click 'Create Account' 4. Enter 'you@example.com' into 'you@example.com' 5. Click 'Create Account'		

Expected: Error message is displayed for incomplete fields

VT-41 | Verify mode switch updates labels on login page [PASSED]

Page: /login	Target: you@example.com	Type: Authentication Tests
1. Navigate to '/login' 2. Enter 'you@example.com' into 'you@example.com' 3. Click 'Sign In' 4. Click 'Dismiss' 5. Enter 'password' into 'Enter password'		

Expected: Labels are updated correctly after mode switch

VT-42 | Check for UI state dances on settings page [PASSED]

Page: /settings

Target: N/A

Type: State Persistence Tests

1. Navigate to '/settings'
2. Click 'Dismiss'
3. Click 'Dismiss'
4. Click 'Dismiss'

Expected: No UI state dances or inconsistencies

VT-43 | Test mode switch with rapid button clicks on about page [PASSED]

Page: /about

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/about'
2. Click 'Dismiss'
3. Click 'Dismiss'
4. Click 'Dismiss'
5. Click 'Dismiss'

Expected: No UI errors or crashes

VT-44 | Verify mode switch retains values on dashboard page [FAILED - BUG FOUND]

Page: /dashboard

Target: N/A

Type: State Persistence Tests

1. Navigate to '/dashboard'
2. Click 'Account v'
3. Click 'Account v'
4. Click 'Dismiss'
5. Click 'Account v'

Expected: Values are retained after mode switch

!! Application did not meet expected behavior - potential bug detected.

VT-45 | Trigger error on contact form submission with empty fields [FAILED - BUG FOUND]

Page: /contact

Target: N/A

Type: Validation Enforcement Tests

1. Navigate to '/contact'
2. Click 'Send Message'

Expected: Error message is displayed

!! Application did not meet expected behavior - potential bug detected.

VT-46 | Verify error recovery on contact form with invalid email [FAILED - BUG FOUND]

Page: /contact

Target: you@example.com

Type: Negative Input Tests

1. Navigate to '/contact'
2. Enter 'invalid' into 'you@example.com'
3. Click 'Send Message'
4. Enter 'valid@example.com' into 'you@example.com'
5. Click 'Send Message'

Expected: Error message is cleared and form submits successfully

!! Application did not meet expected behavior - potential bug detected.

VT-47 | Test error handling on products page with multiple button clic [PASSED]

Page: /products

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/products'
2. Click 'Add Widget Pro'
3. Click 'Add Gadget Lite'
4. Click 'Order'

Expected: Order is processed correctly or an error message is displayed

VT-48 | Verify registration form error handling with missing fields [PASSED]

Page: /register

Target: Jane Doe

Type: Authentication Tests

1. Navigate to '/register'
2. Click 'Create Account'
3. Enter 'Jane Doe' into 'Jane Doe'
4. Click 'Create Account'

Expected: Error message is displayed for missing fields

VT-49 | Check dashboard page error recovery after clicking dismiss [PASSED]

Page: /dashboard

Target: N/A

Type: Authentication Tests

1. Navigate to '/dashboard'
2. Click 'Account v'
3. Click 'Dismiss'

Expected: Dashboard page is restored to its original state

VT-50 | Test login form error handling with empty password field [FAILED - BUG FOUND]

Page: /login

Target: you@example.com

Type: Authentication Tests

1. Navigate to '/login'
2. Enter 'you@example.com' into 'you@example.com'
3. Click 'Sign In'

Expected: Error message is displayed for empty password field

!! Application did not meet expected behavior - potential bug detected.

VT-51 | Verify settings page error handling with dismiss button clic [PASSED]

Page: /settings

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/settings'
2. Click 'Dismiss'

Expected: Settings page is closed or navigated away from

VT-52 | Check about page error recovery after clicking dismiss [PASSED]

Page: /about

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/about'
2. Click 'Dismiss'

Expected: About page is closed or navigated away from

VT-53 | Test contact form error handling with empty name field [FAILED - BUG FOUND]

Page: /contact

Target: you@example.com

Type: Negative Input Tests

1. Navigate to '/contact'
2. Enter 'you@example.com' into 'you@example.com'
3. Enter '+1 (555) 000-0000' into '+1 (555) 000-0000'
4. Click 'Send Message'

Expected: Error message is displayed for empty name field

!! Application did not meet expected behavior - potential bug detected.

VT-54 | Verify products page error handling with multiple add button [PASSED]

Page: /products

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/products'
2. Click 'Add Component X'
3. Click 'Add Gadget Lite'
4. Click 'Add Widget Pro'

Expected: Products are added to the cart or an error message is displayed

7. Structural Validation Confirmation

All 54 test cases were validated against the UI exploration snapshot. The validation engine verified that:

- ✓ All referenced UI element labels exist in the application snapshot.
- ✓ All referenced page paths exist in the crawled page inventory.
- ✓ No hallucinated or invented element labels were detected.
- ✓ Each test case has a valid structure: ID, goal, steps (min 2), and expected result.

8. Appendix

Snapshot Statistics

Metric	Value
Snapshot File	ai_exploration_snapshot.json
Snapshot Size	20912 bytes
Total Pages in Snapshot	9
Total Input Fields	9
Total Buttons	18
Total Navigation Links	65

Pipeline Configuration

Parameter	Value
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AI Model	llama-3.3-70b-versatile (Grog)
Generation Batches	6
Tests per Batch	~10
Batch Delay	30 seconds
Validation Engine	3-Layer Anti-Hallucination
Step Generator	Deterministic (Template-Based)

Output Files Reference

File	Path
Raw Test Cases	data/generated_test_cases.json
Validated Test Cases	data/validated_test_cases.json
Rejected Test Cases	data/rejected_test_cases.json
Gauge Specification	specs/ai_exploration.spec
Step Implementations	step_impl/step_implementation.py
This Report	reports/test_report.pdf